

Solenoid Exact-Triangle Correction

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Summary

Pass 83 corrects the exact-row comparison between the full adelic solenoid

$$\Sigma = (\mathbb{R} \times \widehat{\mathbb{Z}})/\mathbb{Z} \cong \mathbb{A}/\mathbb{Q}$$

and the finite phantom

$$\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}.$$

Projection to the real circle has kernel $\widehat{\mathbb{Z}}$, not ϵ . The phantom appears instead as the quotient of Σ by the dense real line.

Correct Rows

The compact Hausdorff solenoid row is

$$0 \rightarrow \widehat{\mathbb{Z}} \rightarrow \Sigma \rightarrow \mathbb{R}/\mathbb{Z} \rightarrow 0.$$

The finite-prime phantom row is the dense quotient

$$\mathbb{R} \rightarrow \Sigma \rightarrow \epsilon = \widehat{\mathbb{Z}}/\mathbb{Z} \rightarrow 0.$$

Thus ϵ is not a closed subgroup of Σ and should not be called the kernel of $\Sigma \rightarrow \mathbb{R}/\mathbb{Z}$.

Nonsplitting

The compact row does not split continuously. Its Pontryagin dual is

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0,$$

and this row cannot split because any section $\mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}$ would send torsion to torsion in $\mathbb{Q}$, which is torsion-free.

Fourier Boundary

Global characters $\widehat{\Sigma} \cong \mathbb{Q}$ restrict to the closed profinite kernel as

$$\mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}.$$

Only the trivial finite character descends further to $\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$, because descent requires killing the dense diagonal copy of $\mathbb{Z}$. The finite-prime Fourier content is therefore a boundary quotient $\mathbb{Q}/\mathbb{Z}$, not a degree-zero Whittaker or Fourier character of ϵ .

Machine Verification

`code/scripts/check-pass83.py` produced `artifacts/reports/pass83-solenoid-exact-triangle-correction-check.json` with overall PASS. It checks finite dual rows $0 \rightarrow \mathbb{Z} \xrightarrow{\times N} \mathbb{Z} \rightarrow \mathbb{Z}/N \rightarrow 0$, the nonsplitting obstruction for nontrivial N , finite character descent, and the levelwise-zero cokernel that forces ϵ to be read as derived/non-Hausdorff phantom data.

Next Use

The next pass should formulate the derived/solid exact triangle behind $\mathbb{R} \rightarrow \Sigma \rightarrow \epsilon$ and identify the boundary object $\mathbb{Q}/\mathbb{Z}$ as the precise replacement for the missing finite-prime Weyl flip.